

# Pan-India B2B Crop Marketplace

Collective selling, transparent bids and protected payment for small and large farmers.

| PREPARED BY  | PROJECT TYPE                                       | PROPOSED SCOPE                                                            |
|--------------|----------------------------------------------------|---------------------------------------------------------------------------|
| Cyber Shield | B2B web/mobile application<br>AI help chatbot only | Crops across India; individual and FPO sellers;<br>banking partner escrow |

## 1. Detailed Problem Statement

### Background

Farmers bear cultivation costs and harvest risk, but often depend on a few intermediaries to reach companies, processors, institutions and retail stores. Delayed collection can weaken their bargaining power, especially when crops are perishable. Small farmers also struggle to supply the quantities that bulk buyers require.

### Illustrative tomato scenario

A farmer expects about INR 30 per kg for tomatoes. Collection is delayed until the crop is close to over-ripening, and the offered price falls to INR 5-10 per kg. The downstream selling price may remain much higher. These figures are illustrative; they show how delay and limited buyer access can lead to a distress sale.

### Core problems

- Small farmers lack an easy way to combine their produce into a verified bulk lot.
- Small and large farmers need clear buyer offers, dependable pickup and traceable payment.
- Buyers need verified farmer, crop, quantity, harvest and collection information.
- Farmers need help with registration, officer contacts, complaints and simple portal instructions.

#### PROJECT OBJECTIVE

**Connect verified individual farmers and FPO groups with registered buyers across India, secure buyer payment before pickup, and initiate seller settlement after verified handover.**

This is an SIH proposal and prototype scope. Central Government administration and live banking integration are proposed arrangements, not claims of an existing deployment.

## 2. Proposed Solution

### One marketplace for small farmers and bulk suppliers

The proposed platform supports small farmers through group selling and allows large or bulk-capable farmers to sell individually. A **Farmer Producer Organisation (FPO)** can combine members' crops into a larger supply lot. The Aavin milk-society analogy explains collective supply: members bring their produce together, sell collectively and receive their respective earnings.

The proposed Central Government role is to administer the application, policies and authorised users. Agricultural officers manage field verification and assisted registration. A **banking partner operates escrow and settlement**. Companies, processors, colleges, institutions and retail stores participate as registered buyers.

### Operating model

| FUNCTION           | HOW IT WORKS                                                                                                                                                     |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inclusive supply   | Verified small farmers contribute to an FPO lot; large farmers can offer their own lots. Minimum landholding is not a blanket seller-entry condition.            |
| Officer assistance | An agricultural officer visits the land, collects details, uploads the record and sends the farmer a secure access link.                                         |
| Open bidding       | Buyers submit quantity-specific offers in kg during a short, published bidding window. The farmer selects preferred offers; the officer confirms with the buyer. |
| Protected payment  | Each confirmed order is funded through banking partner escrow before goods are released. Verified handover triggers seller settlement.                           |
| Buyer logistics    | The buyer arranges its transportation partner, vehicle and collecting person. Their details are linked to the order.                                             |
| Accessible support | District and taluka officer contacts, grievances and a simple help chatbot are available to farmers.                                                             |

### Produce, payment and control

- **Produce:** individual farmer or FPO collection point -> buyer's authorised vehicle.
- **Payment:** buyer -> banking partner escrow -> verified seller account. FPO member distribution follows a separately agreed arrangement.
- **Control:** field verification -> farmer selection -> officer and buyer confirmation -> recorded handover.

#### FIRST-VERSION SCOPE

**Crops only. Both small and large farmers are included. One lot can serve several buyers through separate orders. AI is limited to the help chatbot.**

FPO terminology: [Small Farmers' Agri-Business Consortium - Farmer Producer Organization Scheme](#). The Aavin comparison is an analogy for collective supply; detailed FPO payment rules are still to be agreed.

### 3. Roles, Permissions and Data Access

The following is the proposed minimum permission model. Farmers retain control over their preferred buyers and quantities. Agricultural officers have broader registration and verification access; additional farmer-editable fields will be finalised later.

| ROLE                          | MAIN RESPONSIBILITIES                                                                                                                               | PERMISSION BOUNDARY                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Farmer                        | Use the access link; view records, crop listings and bids; choose offers and quantities; request corrections; raise complaints.                     | Cannot approve own verification or independently mark funds as paid. Extra editable fields remain undecided.  |
| FPO representative            | Coordinate verified members, record each contribution and offer an aggregated lot with member agreement.                                            | Cannot silently change member contributions or the agreed distribution arrangement.                           |
| Agricultural officer          | Visit land; collect and upload farmer, crop and initial bank details; verify field facts; manage bidding deadlines; confirm allocations and pickup. | Initial bank-data capture is separate from payout control. Cannot receive buyer funds or redirect settlement. |
| Buyer / vendor                | Complete registration; bid for a stated kg quantity; confirm orders; fund escrow; assign transport and accept custody.                              | Can order from verified sellers only. Farmer contact access follows the agreed disclosure stage.              |
| Government administrator      | Manage crop catalogue, buyer verification, authorised officers, jurisdictions, support contacts, policy settings and audits.                        | Administration does not include holding escrow funds or silently editing completed records.                   |
| Banking partner               | Confirm funds received; hold order-linked balances; process authorised settlement and refunds.                                                      | Releases follow the agreed order and dispute rules. The prototype simulates these events.                     |
| Authorised driver / collector | Present order-linked identity and OTP or QR; collect the accepted load and confirm custody.                                                         | Cannot change the buyer, price, beneficiary or order allocation.                                              |

#### Data visibility

- **Marketplace:** crop, variety, kg quantity, harvest date, approximate location, price and published bid details.
- **Officer access:** farmer and buyer phone numbers, land records, inspection notes and supporting documents.
- **Buyer contact access:** baseline proposal is after order confirmation. Earlier access immediately after bidding remains an open decision.
- **Financial identity:** bank details are masked in routine views. Later changes require farmer confirmation and independent verification; every change is logged.

#### FARMER ACCESS

**Send a secure activation link to the registered mobile number. After the farmer verifies access, show their own records and the actions allowed for their role.**

## 4. Setup and Farmer Registration

### Phase 1 - Platform setup

| STEP | ACTION                                                                                                                                          | RESULT                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 1    | Administrator prepares the crop and variety catalogue, kg unit, reference-price fields and location hierarchy.                                  | Consistent listing choices.            |
| 2    | Configure buyer categories, document checklists, bidding and cancellation settings, officer jurisdictions and support contacts.                 | Published operating rules.             |
| 3    | Assign authorised officers and backup coverage. Configure the proposed banking partner escrow workflow; label prototype payment events as demo. | Responsible owners and payment states. |

### Phase 2 - Officer-assisted farmer onboarding

| STEP | ACTION                                                                                                                                                     | RESULT                                   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 4    | Farmer requests registration or approaches the local agricultural officer.                                                                                 | Onboarding request.                      |
| 5    | Officer visits the farmer's land and checks the farmer, location and crop information.                                                                     | Field visit record.                      |
| 6    | Collect identity/contact details, land address and map location, ownership or valid lease evidence, crop/variety, expected yield and initial bank details. | Farmer and land data.                    |
| 7    | Officer uploads the information, supporting documents, inspection notes and photographs to the portal.                                                     | Submitted seller record.                 |
| 8    | Complete the required checks; record approval or corrections. Initial bank details follow the approved verification process.                               | Verified or correction-required profile. |
| 9    | Send the farmer a secure access link. Farmer views the recorded details, requests corrections and sees officer contacts and support.                       | Farmer portal access.                    |

### How FPO aggregation is recorded

- Link the FPO or approved group representative to its participating, verified farmers.
- Record each member's crop, variety, expected kg contribution and harvest window.
- Combine compatible produce into a group lot and prevent the same quantity being offered twice.
- Keep member contribution and sale records. The route and formula for member payments and any profit sharing remain to be finalised.

#### INCLUSIVE ELIGIBILITY

**Verify land and production capacity without excluding small farmers by land size. Any bulk-order threshold applies to the offered lot, including a combined FPO lot.**

## 5. Buyer Registration and Farmer Support

### Phase 2 continued - Register and verify buyers

| STEP | ACTION                                                                                                                             | RESULT                          |
|------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 10   | Buyer selects its category and uploads one or two relevant verification documents.                                                 | Category-based application.     |
| 11   | Record organisation/store name, address, authorised purchaser, mobile number and delivery location. Confirm the contact number.    | Reachable buyer representative. |
| 12   | Administrator or assigned verification officer reviews the documents, seeks corrections if needed and activates the buyer account. | Verified buyer access.          |

### Short document checklists

| BUYER CATEGORY        | DOCUMENT 1                                       | DOCUMENT 2, IF NEEDED                              |
|-----------------------|--------------------------------------------------|----------------------------------------------------|
| Company / processor   | Business registration proof.                     | Authorisation for the purchasing representative.   |
| College / institution | Institutional registration or recognition proof. | Authorisation for the purchasing representative.   |
| Retail store          | Shop or business registration proof.             | Proprietor or authorised purchaser identification. |

These are proposed portal verification examples, not a statement of statutory requirements. The final accepted document list will be agreed before implementation.

### Officer directory available from the beginning

The farmer dashboard shows the assigned agricultural officer, relevant taluka/block officer and district agricultural officer, with office/jurisdiction, mobile or official contact number, and available support details. Directory entries are maintained by authorised administrators.

### Grievance mechanism

- Farmer or buyer selects a complaint category, identifies the listing/order if relevant, describes the issue and adds supporting evidence.
- The portal provides a complaint number, assigned officer, status and response history.
- The assigned officer responds or escalates through the configured taluka/district support hierarchy. Closure records the outcome and reason.

### Simple chatbot at the front of the portal

The chatbot answers short questions such as "How do I open my farmer profile?", "How do I choose a bid?", "Who is my officer?" and "How do I file a complaint?" It uses approved portal instructions and points users to the relevant screen or officer. It does not choose buyers, decide disputes, change records or release payment.

#### SUPPORT STAYS HUMAN

**Use the chatbot for instructions. Use authorised officers for verification, bid confirmation and complaint resolution.**

## 6. Crop Listings and India-wide Discovery

### Phase 3 - Supply listings and buyer discovery

| STEP | ACTION                                                                                                                                         | RESULT                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 13   | Choose crop and variety from the catalogue dropdowns. The officer assists with the initial entry.                                              | Standard crop identity.               |
| 14   | Enter expected or available quantity in kg, crop images, expected harvest date, pickup window and minimum price per kg.                        | Complete supply information.          |
| 15   | Review the listing against the verified farmer/FPO record and publish it when ready. Record relevant quality details through human inspection. | Verified crop listing.                |
| 16   | Buyer browses all-India listings or applies region, state and district filters, together with crop and availability filters.                   | Matching listings by selected fields. |
| 17   | Buyer opens a listing to review supply details and bid, or posts a quantity-specific demand request.                                           | Buyer interest or open demand.        |

### Catalogue and kg convention

Build a comprehensive, maintainable catalogue covering cereals, pulses, vegetables, fruits and other supported crops, including rice, wheat, bananas and coconuts. Each crop has relevant varieties. If a variety is missing, the officer requests a catalogue update. The catalogue is structured data; it is not an AI training requirement.

| FIELD                  | PORTAL RULE                                                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crop and variety       | Use searchable dropdowns. Identify the actual product form where needed, such as paddy versus milled rice.                                                               |
| Quantity and price     | List all quantities in kg and unit prices in INR per kg. For bananas or coconuts, use weight rather than silently treating a bunch or piece as a kg.                     |
| Yield and harvest      | Label expected yield as an estimate. Show expected harvest date and the agreed pickup window; actual handover weight is recorded later.                                  |
| Source and destination | Show source region/state/district and pickup area. Record buyer delivery state/district. The exact seller-selected destination-coverage feature remains to be finalised. |

### Location browsing

| SELECTION                 | RESULT                                                                                                   |
|---------------------------|----------------------------------------------------------------------------------------------------------|
| No location filter        | Show eligible farmer and crop listings across India.                                                     |
| Region or state selected  | Narrow results to that area. Region is optional; a buyer may select a state directly.                    |
| State + district selected | Show farmers and crop listings in that district, for example Tamil Nadu followed by a selected district. |

#### HUMAN DECISIONS AND ORDINARY FILTERS

The portal filters records using selected fields. AI does not compare crop timing, grade, quality, quantity or bids to recommend a winner.

## 7. Transparent Bidding and Allocation

### Phase 4 - Short bidding window and farmer choice

| STEP | ACTION                                                                                                                                                               | RESULT                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 18   | Officer sets a short opening and closing window. Display the deadline clearly to participating buyers.                                                               | Published bidding window.     |
| 19   | Buyer states requested kg, offered price per kg, total amount, proposed pickup date and bid validity.                                                                | Quantity-specific offer.      |
| 20   | Display comparable bid prices, quantities, totals and timestamps to eligible participants. Show the reference price with source/date and the farmer's minimum price. | Transparent comparison.       |
| 21   | Farmer selects preferred offers through the access link and proposes how many kg to allocate to each buyer.                                                          | Farmer's recorded choices.    |
| 22   | Officer reviews the farmer's choices and calls the buyer to confirm quantity, price and pickup arrangements.                                                         | Officer confirmation record.  |
| 23   | Buyer confirms the final terms. Reserve quantities without exceeding the lot balance; keep confirmation traceable in the portal.                                     | Confirmed allocations.        |
| 24   | Create a separate order for each accepted allocation. Any remaining quantity stays available.                                                                        | Independent order lifecycles. |

### Partial procurement is part of the bid

A buyer bids for the quantity it wants, rather than buying the whole lot. For a 50 kg listing, the following illustrative bids can both be accepted:

| BUYER             | REQUESTED KG | PRICE / KG | ORDER VALUE |
|-------------------|--------------|------------|-------------|
| Company A         | 10 kg        | INR 20     | INR 200     |
| Institution B     | 20 kg        | INR 22     | INR 440     |
| Total allocated   | 30 kg        | -          | INR 640     |
| Balance available | 20 kg        | -          | -           |

The farmer may prefer different buyers. If the farmer proposes a quantity different from a bid, the buyer confirms the revised quantity and price before the order is created. Pending reservations must be released if confirmation or payment expires.

#### NO AUTOMATIC WINNER

**The farmer chooses; the officer verifies and confirms with the buyer. Each order is priced for its confirmed quantity before escrow funding. A later shortage follows the exception policy.**

Open setting: the exact bidding duration and its time unit are still to be decided. No fixed number of minutes, hours or days is assumed.

## 8. Order Funding and Buyer-arranged Pickup

### Phase 5 - Order agreement and banking partner escrow

| STEP | ACTION                                                                                                                                                                        | RESULT                    |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 25   | Snapshot the agreed crop/variety, kg quantity, unit price, applicable quality terms, pickup deadline, cancellation terms and custody rule.                                    | Digital order agreement.  |
| 26   | Generate an order-specific payment request for confirmed quantity multiplied by agreed price per kg.                                                                          | Awaiting Payment status.  |
| 27   | Buyer prepays the full order amount into banking partner escrow. The prototype demonstrates this using clearly labelled simulated payment events.                             | Order-linked held funds.  |
| 28   | Mark payment secured only after the bank receipt is confirmed, or after the explicit demo event in the prototype. If payment expires, cancel the order and free its quantity. | Secured or expired order. |

#### PAYMENT DIRECTION

**Buyer -> banking partner escrow -> verified seller account after verified pickup. Buyer refunds return through the banking partner. Officers do not receive or hold the money.**

### Phase 6 - Transport selected by the buyer

| STEP | ACTION                                                                                                                                                 | RESULT                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 29   | Buyer registers its transportation partner, assigned vehicle number/capacity, driver or collecting person, and their contact number against the order. | Traceable transport assignment. |
| 30   | Confirm pickup location, arrival time and the crop quantity to prepare. Share the relevant confirmed-order contacts for coordination.                  | Pickup-ready arrangement.       |
| 31   | Assign the agricultural officer or authorised inspector for handover. Reassign to another authorised officer if necessary.                             | Responsible pickup verifier.    |

### Simple collection authentication

- Use an order-linked OTP or QR challenge to check that the arriving person is authorised for that order.
- Match the driver/collector and vehicle against the recorded assignment. Changes require an updated, auditable assignment.
- Treat OTP/QR confirmation as identity and event evidence, not as proof of crop weight or quality.
- The buyer arranges and pays for transport. Route coordination can be manual; no AI route recommendation is included.

The buyer's access to the farmer's phone number is proposed to begin at order confirmation. Earlier disclosure at bid submission remains an open decision; contact details are not part of the public bidding table.

## 9. Verified Handover and Settlement

### Phase 7 - Check the goods and confirm custody

| STEP | ACTION                                                                                                                                                        | RESULT                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 32   | Driver or collector presents the order-linked OTP or QR and the recorded vehicle/identity details at the pickup point.                                        | Authorised arrival.            |
| 33   | Weigh the crop in kg and check it against the agreed crop, variety and applicable quality terms. Record any shortage or mismatch before handover approval.    | Accepted or disputed quantity. |
| 34   | Attach the weight record, crop photographs, time, location and officer inspection notes to the order.                                                         | Pickup evidence.               |
| 35   | Farmer or authorised FPO representative, collecting person and officer confirm the accepted handover. Mark Dispatch Verified; custody transfers to the buyer. | Verified dispatch receipt.     |

### Phase 8 - Seller payment and order closure

| STEP | ACTION                                                                                                                                                         | RESULT                               |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 36   | Confirm the payable value from the accepted quantity and agreed unit price. If the full order is accepted, the full agreed order value is payable.             | Verified settlement amount.          |
| 37   | Banking partner releases the authorised amount to the verified seller account after the required pickup confirmations. Mark Paid only after bank confirmation. | Seller payment and receipt.          |
| 38   | Where a documented shortage, rejection or cancellation changes the payable amount, process the agreed refund or hold under the dispute policy.                 | Traceable refund or dispute balance. |
| 39   | For an FPO sale, record the group receipt against the order and member contributions. Apply the separately agreed member-payment arrangement.                  | Transparent group sale record.       |
| 40   | Buyer transports the goods, confirms arrival and closes the completed order. Preserve the dispatch, payment and complaint history.                             | Delivered and Closed status.         |

#### FULL SETTLEMENT FOLLOWS VERIFIED PICKUP

**The buyer selected the transport and its authorised collector accepted custody. Ordinary transport responsibility moves to the buyer at verified handover; seller payment is triggered at pickup.**

### Quantity differences are exceptions

Partial procurement is already fixed in the accepted bid. A later weight or quality difference is recorded as an exception and resolved before the affected funds are released. Any accepted, undisputed portion can be handled under the agreed policy; no party silently rewrites the order.

Actual bank processing time depends on the banking arrangement. The prototype shows settlement states and receipts, without claiming that a simulated transfer is a live payment.

## 10. Cancellation and Dispute Policy

The portal needs clear rules for bid withdrawal, order cancellation, late cancellation and disputes. The following is the proposed policy structure; exact deadlines, charges and appeal periods remain to be agreed.

### Cancellation depends on the stage

| STAGE                             | PROPOSED HANDLING                                                                                                                     | STOCK AND PAYMENT                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Bid not yet accepted              | Allow withdrawal or amendment within the published bidding rules; retain the history.                                                 | No confirmed order payment. Release any temporary hold.                                         |
| Order confirmed; unpaid           | Record cancellation or payment expiry and notify the other parties. Apply the published stage rule.                                   | Cancel the order and return available quantity to sale.                                         |
| Payment secured; before pickup    | Require a reason and recorded decision. Bank processes the authorised refund and any agreed compensation.                             | Reopen the uncollected quantity when still available.                                           |
| Late buyer cancellation / no-show | Apply a published compensation payment to the farmer for a buyer-caused late cancellation. Officer records evidence and the decision. | Bank pays the agreed compensation to the farmer and refunds the remaining amount as applicable. |
| After verified handover           | Handle through the dispute process; this is not an ordinary uncollected-order cancellation.                                           | Do not relist goods already dispatched. Review any claim using handover evidence.               |

### Other exception paths

| ISSUE                             | CONTROL AND RESOLUTION                                                                                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Short quantity / quality mismatch | Record the difference, pause settlement of the disputed portion and seek an accepted resolution or escalation.     |
| Pickup delay                      | Record the delay; use the published grace period, rescheduling, cancellation and compensation rules.               |
| Unapproved vehicle / collector    | Pause handover until an authorised reassignment is recorded.                                                       |
| Officer unavailable               | Reassign to another authorised officer; administrators can add approved officers and local coverage.               |
| Fraud or side-payment complaint   | Preserve evidence and refer the case to the authorised reviewer; payment holds follow the banking dispute process. |

### Dispute handling

Complaint opened -> evidence collected -> assigned officer review -> district/taluka escalation where needed -> recorded decision -> authorised settlement/refund action -> closure or appeal. The concerned parties can view the status and outcome.

#### POLICY VALUES STILL OPEN

**Bidding and cancellation cutoffs, late-cancellation compensation, dispute response times and appeal periods remain to be agreed and published before live orders are accepted.**

# 11. Status Model and Prototype Scope

## Simple status lifecycles

| RECORD              | STATUS SEQUENCE                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Farmer / FPO member | Onboarding Requested -> Field Visit -> Submitted -> Verified / Correction Required                                            |
| Buyer               | Submitted -> Under Review -> Verified / Correction Required                                                                   |
| Listing             | Draft -> Open for Bids -> Partially Allocated -> Fully Allocated / Expired / Cancelled                                        |
| Order               | Awaiting Confirmation -> Awaiting Payment -> Secured -> Vehicle Assigned -> Inspection -> Dispatch Verified -> Paid -> Closed |
| Escrow              | Awaiting -> Received -> Held -> Released / Partial Refund / Full Refund / Disputed                                            |
| Grievance / dispute | Opened -> Assigned -> Under Review -> Escalated if needed -> Resolved -> Closed / Appealed                                    |

## Implementation milestones

| MILESTONE                  | MINIMUM DEMONSTRATION                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1. Access and onboarding   | Farmer, FPO, buyer and officer roles; officer field record; secure farmer link; short buyer verification checklist.               |
| 2. Catalogue and listings  | Crop/variety dropdowns; kg quantity; images and harvest date; FPO contributions; all-India and state/district search.             |
| 3. Bids and confirmation   | Published short deadline; quantity and unit-price bids; farmer allocation; officer phone-confirmation record; buyer confirmation. |
| 4. Demo banking escrow     | Separate order value; simulated funding, held balance, release/refund and receipt; clear demo labels.                             |
| 5. Pickup and closure      | Registered transport/collector; OTP or QR; weight and inspection evidence; confirmed handover; payment states and closure.        |
| 6. Help and accountability | Officer directory; grievance tracking; instruction chatbot; cancellation paths; timestamped action history.                       |

## Completion checks

- A verified small farmer can participate through an FPO lot; an individual bulk farmer can list directly.
- A 50 kg lot can produce separate 10 kg and 20 kg orders while keeping 20 kg available.
- Pickup cannot proceed as paid unless the order has a secured-payment record; Paid requires a settlement confirmation.
- The farmer chooses bids, the officer records confirmation, and the chatbot only provides help.
- Every approval, edit, allocation, payment transition and complaint decision records actor, time and reason.

## 12. Limits, Future Scope and Open Decisions

### Current scope and practical limits

- The current proposal covers B2B crops, small and large farmers, FPO aggregation and pan-India discovery. Direct consumer sales, dairy, eggs and livestock remain outside this version.
- The SIH prototype simulates banking partner escrow. A live service needs an agreed banking integration and operational arrangements for verification and disputes.
- Land mapping supports field verification; a map pin alone does not prove ownership. Expected yield remains an estimate until the crop is weighed.
- Officers assist during connectivity interruptions. Detailed offline recovery is outside the current prototype; payment and verification confirmations still need connectivity.
- Officer workload requires jurisdiction assignment, backup coverage and additional authorised personnel. The proposal does not assume unlimited officer capacity.

### Future scope - keep these as later integrations

| CAPABILITY                    | LATER IMPLEMENTATION                                                                                                           |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Live banking partner escrow   | Bank-confirmed payments, settlement, refunds, reconciliation and production dispute handling.                                  |
| Official land-record services | Authorised data connections with correction and officer-review routes.                                                         |
| Live reference prices         | Approved market-price sources with timestamps; prices provide context rather than an AI buying decision.                       |
| Regional languages and access | Additional Indian languages, voice guidance and accessibility improvements. The AI role remains the help chatbot.              |
| Offline support               | Capture field notes when disconnected and synchronise when service returns; do not treat offline entries as bank confirmation. |
| Transport integrations        | Optional links to buyer transportation partners and shipment updates; buyer responsibility remains unchanged.                  |

### Decisions still to be finalised

Exact bidding duration; extra farmer-editable fields; the final buyer-document list; FPO member-payment and profit-sharing rules; contact disclosure at bid submission versus order confirmation; destination-state selection; cancellation compensation and dispute deadlines. An additional participant category also needs confirmation: transport/loading workers or exporters.

#### REVISED PROJECT PRINCIPLE

**Make small-farmer participation practical, keep buyer offers visible, preserve farmer choice, support decisions through officers, and protect seller payment through a banking partner.**

All platform functions and integrations described here are proposed for the SIH prototype or future implementation, as labelled.